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Number System

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Objectives

- Number System.
- Binary Arithmetic

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Introduction

- Digital systems including computers run on a 2-valued system called **binary system**.
- Transistors are the basic elements in digital systems with ON and OFF states.
- The binary system has only two digits 0 (represents OFF) and 1(represents ON) called **bi**nary digi**t** or **bit**.
- Computer translates everything into binary numbers as they can understand only binary numbers.

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Number Systems

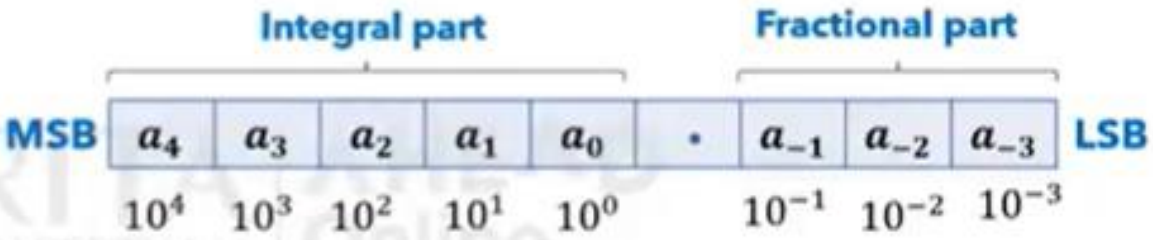
- Number system is the method to represent numbers.
- The base of each number system is called radix which determines how many different symbols are needed to represent a number.

Number System	Base	Digits
Decimal	10	0,1,2,3,4,5,6,7,8,9
Binary	2	0,1
Octal	8	0,1,2,3,4,5,6,7
Hexadecimal	16	0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

Decimal Number System

- Referred to as base-10 with ten digits 0,1,2,3,4,5,6,7,8,9
- Number representation

$$a_4\, a_3\, a_2\, a_1\, a_0\, .\, a_{-1} a_{-2} a_{-3}$$



$$= 10^4 a_4 + 10^3 a_3 + 10^2 a_2 + 10^1 a_1 + 10^0 a_0 + 10^{-1} a_{-1} + 10^{-2} a_{-2} + 10^{-3} a_{-3}$$

e.g., $(765)_{10} = 7 \times 10^2 + 6 \times 10^1 + 5 \times 10^0$

$= 7 \times 100 + 6 \times 10 + 5 \times 1$

$= 765$

Binary Number System

- Referred to as base-2 with two digits 0,1.
- Coefficient 'a_j' is the digit between 0 to base-1 i.e., in binary coefficient are a_j = 0 and 1.

MSB								LSB	
a_4	a_3	a_2	a_1	a_0	•	a_{-1}	a_{-2}	a_{-3}	
2^4	2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}	Power of base
16	8	4	2	1		0.5	0.25	0.125	

- Example : $(11011.11)_2$, $(1001)_2$
- $(1)_{10} = (1)_2$, $(2)_{10} = (10)_2$, $(16)_{10} = (10000)_2$, $(22)_{10} = (10110)_2$

Octal Number System

- Base-8 number system with eight digit to represent numbers
- The coefficients are one of the eight digits (0,1,2….,7) i.e., between 0 to (8-1)

MSB								LSB	
a_4	a_3	a_2	a_1	a_0	.	a_{-1}	a_{-2}	a_{-3}	
8^4	8^3	8^2	8^1	8^0		8^{-1}	8^{-2}	8^{-3}	Power of base
4096	512	64	8	1		0.125	0.0156		

- Example : $(2765)_8$, $(1435)_8$
- $(7)_{10} = (7)_8$, $(8)_{10} = (10)_8$, $(128)_{10} = (200)_8$

Hexadecimal Number System

- Base-16 number system with 16 digits for number representation.
- First ten digits are borrowed from the decimal system and letters A, B, C, D, E, F are used for digits 10, 11, 12, 13, 14, 15.

MSB				LSB	
a_4	a_3	a_2	a_1	a_0	
16^4	16^3	16^2	16^1	16^0	**Power of base**

- Example : $(C74F)_{16}$, $(93AC)_{16}$

Counting in Number System

Decimal	Binary Number	Octal	Hexadecimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C

Why binary not decimal?

- Transistors are the fundamental unit in digital computer with only 2 states ON and OFF.
- In decimal system : 4 transistor can represent only maximum 4 numbers by keeping all ON.
- With binary : 4 transistor can represent any number up to 15 i.e., we can represent up to 16 values from 0 to 15.

Binary Operations

- **Addition**

0 + 0 = 0
0+ 1 = 1
1+ 1 = 0 carry 1

Example:

```
      1 1 1 1
    1 0 1 1 0 1
  + 1 0 0 1 1 1
  -------------
  1 0 1 0 1 0 0
```

Subtraction

0-0 = 1-1 = 0
1-0 = 1
0-1 = 1 borrow 1

Example:

```
      0->10 1
    1 0 1 1 0 1
  - 1 0 0 1 1 1
  -------------
    0 0 0 1 1 0
```

Multiplication

```
      1 0 1 1
  X     1 0 1
  -----------
      1 0 1 1
    0 0 0 0
  1 0 1 1
  -----------
  1 1 0 1 1 1
```

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Binary Division

Input A	Input B	Divide (D) A/B
0	0	Not defined
0	1	0
1	0	Not defined
1	1	1

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Binary Division

```
        10011
101 | 101101
       101
       ----
       0001
          0
          --
          10
          00
          --
          101
          101
          ---
          000
```

5

32 8 4 1

101101 = 45

9 → 1001

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Objectives

- Number Base Conversions.

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Number base conversions

- **Converting from any number system to decimal system**

- If the base 'r' and the coefficients are given, then any number can be converted to decimal equivalent by multiplying each coefficient with the corresponding power of r and adding.

- $a_n * r^n + a_{n-1} * r^{n-1} + \ldots\ldots + a_1 * r^1 + a_0 + a_{-1} * r^{-1} \ldots\ldots a_{-m} * r^{-m}$

Example

2nd position - 2^2

- $(1101)_2$ to decimal

0th position - 2^0

$$= 1*2^3 + 1*2^2 + 0*2^1 + 1*2^0$$
$$= 8 + 4 + 0 + 1 = (13)_{10}$$

- $(630.4)_8$ to decimal

1st position - 8^1

$$= 6*8^2 + 3*8^1 + 0*8^0 + 4*8^{-1}$$
$$= 384 + 24 + 0 + 0.5 = (408.5)_{10}$$

- $(C76F)_{16}$ to decimal

$$= 12*16^3 + 7*16^2 + 6*16^1 + 15*16^0$$
$$= 49{,}152 + 1{,}792 + 96 + 15 = (51055)_{10}$$

Number base conversions

➢ Convert decimal numbers to other number systems.

- Divide the decimal number repeatedly with the base until no longer possible to divide.
- Divide the quotient and note the remainder.
- Remainders of the division put together in reverse order is the result.
- For fractional part, multiply the decimal fraction with base repeatedly until we reach zero or close to zero. The integer part together is the result of decimal fraction.

$(13.125)_{10}$ to binary

	Q	R
13/**2** =	6	**1**
6/**2** =	3	**0**
3/**2** =	1	**1**
1/**2** =	0	**1**

$(13)_{10}$ = 1 1 0 1

0.125 * 2 = **0**. 25

0.25 * 2 = **0**. 5

0.5 * 2 = **1**. 0

$(0.125)_{10}$ = 001

$(13.125)_{10}$ = 1101.001

Example

- $(25.13)_{10}$ to binary

2/2 quotient = 12 ; remainder = **1**
12/2 quo = 6 ; rem = **0**
6/2 quo = 3 ; rem = **0**
3/2 quo = 1 ; rem = **1**
1/2 quo = 0 ; rem = **1**

Fractional Part binary equivalent
0.13 * 2 = **0**.26
0.26 *2 = **0**.52
0.52 *2 = **1**.04

$(25.13)_{10}$ = 11001 + .001
= $(11001.001)_2$

$(153)_{10}$ to octal

153/8 quo = 19 ; rem = **1**
19/8 quo = 2 ; rem = **3**
2/8 quo = 0 ; rem = **2**

$(153)_{10} = (231)_8$

$(2545)_{10}$ to hexadecimal

2545/16 quo = 159 ; rem = **1**
159/16 quo = 9 ; rem = 15 (**F**)
9/16 quo = 0; rem = **9**

$(2545)_{10} = (9F1)_{16}$

Number base conversions

➢ Octal and Hexadecimal numbers

- Octal digit corresponds to three binary digit (111 = 7 is the maximum range of octal coefficient)
- $(256)_8$ to binary = 2 5 6 = $(010\ 101\ 110)_2$

  010 101 110
- $(1110.0101)_2$ to octal = 001 110 . 010 1 = $(16.21)_8$

  1 6 2 1

Number base conversions

- Hexadecimal digit corresponds to four binary digits.
- Binary to hexadecimal is by partitioning the binary number into groups of four digits each.
- $(\underline{10}\ \underline{1100}\ \underline{0110}\ \underline{1011}.\underline{1111}\ \underline{0010})_2 = (1C6B.F2)_{16}$

  1 C 6 B F 2
- $(306.D)_{16} =$ 3 0 6 . D

  0011 0000 0110 1101

Reference

- M Morris Mano - Computer System Architecture - PHI - Third Edition
- Gideon Langholz, Abraha& Joe L Mott - Digital Logic Design - World Scientific Publishing Co Ltd
- Thomas C Bartee - Digital Computer Fundamentals - Tata Mc Graw Hill - Sixth Edition

Objectives

- Binary Coding Schemes.

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Binary Coding

- Coding is a technique by which a set of objects is represented in a machine by a set of binary numbers called codes.
- A process of coding is required where different data is coded in a unique combination of 0s and 1s using coding schemes.
- Data is coded with binary digits.
- Binary coding can be done with numbers or alphanumeric letters.

Binary Coding Schemes

Many coding schemes.

- Weighted Code.
- Non Weighted Code.
- Alpha Numeric Codes.
- Error correcting and Detecting Codes.

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Weighted Codes

- Follow the positional weighting principle.
- Specific weight assigned to each position.
- Eg. 8421 code/BCD code.
  - 8, 4, 2 and 1 are the weights of the 4 bits of the BCD code.
  - The weight of LSB is 2^0 or 1, that of the next higher order bit is 2^1 or 2, that of the next higher order bit is 2^2 or 4 and that of MSB is 2^4 or 8.

8 4 2 1

BCD Code

- Each decimal number is converted to its four bit binary equivalent.

638

0110 0011 1000

3 = 0×8 + 0×4 + 1×2 + 1×1

= 3

- Weight assigned are 8-4-2-1.
- Here bit assignment for 6 is 0110 – 0*8+1*4+1*2+0*~~6~~1

= 6

Non Weighted Code

- No positional weight.

Excess - 3 code.

Grey code.

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Excess – 3 Code

- Add 3 to decimal number and then convert in to binary form.

Excess- 3 code of 5

5 + 3 = 8

Binary of 8 = 1000

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Grey Code

- A non weighted code.
- Also known as minimum change code.
- Two successive values differ by only one bit.

Decimal	Grey
0	0000
1	0001
2	0011
3	0010
4	0110
5	0111

Objectives

- Alpha Numeric Codes.
- Error correction and detecting codes.

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Alpha Numeric Codes

- A complete alphanumeric code would include the 26 lowercase characters, 26 uppercase characters, 10 numeric digits, etc.
- Many alpha numeric codes are there.

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ASCII

- **ASCII** (American Standard for Information Interchange) is a coding system for representing characters.
- It uses 8 bits to represent a character.
- The MSB is set as the parity bit. So it can represent only 128 characters.
- Extended ASCII discards the parity bit and it can represent 256 characters.

ASCII Codes

Symbol	ASCII	Hex	Symbol	ASCII	Hex	Symbol	ASCII	Hex
line feed	00001010	0A	>	00111110	3E	^	01011110	5E
carriage return	00001011	0B	?	00111111	3F	_	01011111	5F
space	00100000	20	@	01000000	40	`	01100000	60
!	00100001	21	A	01000001	41	a	01100001	61
"	00100010	22	B	01000010	42	b	01100010	62
#	00100011	23	C	01000011	43	c	01100011	63
$	00100100	24	D	01000100	44	d	01100100	64
%	00100101	25	E	01000101	45	e	01100101	65
&	00100110	26	F	01000110	46	f	01100110	66
'	00100111	27	G	01000111	47	g	01100111	67
(	00101000	28	H	01001000	48	h	01101000	68
)	00101001	29	I	01001001	49	i	01101001	69
*	00101010	2A	J	01001010	4A	j	01101010	6A
+	00101011	2B	K	01001011	4B	k	01101011	6B
,	00101100	2C	L	01001100	4C	l	01101100	6C
-	00101101	2D	M	01001101	4D	m	01101101	6D
.	00101110	2E	N	01001110	4E	n	01101110	6E
/	00111111	2F	O	01001111	4F	o	01101111	6F
0	00110000	30	P	01010000	50	p	01110000	70
1	00110001	31	Q	01010001	51	q	01110001	71
2	00110010	32	R	01010010	52	r	01110010	72
3	00110011	33	S	01010011	53	s	01110011	73
4	00110100	34	T	01010100	54	t	01110100	74
5	00110101	35	U	01010101	55	u	01110101	75
6	00110110	36	V	01010110	56	v	01110110	76
7	00110111	37	W	01010111	57	w	01110111	77
8	00111000	38	X	01011000	58	x	01111000	78
9	00111001	39	Y	01011001	59	y	01111001	79
:	00111010	3A	Z	01011010	5A	z	01111010	7A
;	00111011	3B	[	01011011	5B	{	01111011	7B
<	00111100	3C	\	01011100	5C	|	01111100	7C
=	00111101	3D	]	01011101	5D	}	01111101	7D

ASCII Table

- ASCII value of number 0 :- 48
  - Numeric range → 48-57
- ASCII value of capital letter A :- 65
  - Alpha range (Uppercase Alphabets) → 65-90
- ASCII value of small letter a :- 97
  - Alpha range (Lowercase Alphabets) → 97-122
- ASCII value of space :- 32

EBCDIC

- "Extended Binary Coded Decimal Interchange Code"
- Used in IBM equipments and in large computers for communicating alphanumeric data.
- 8 bit code and the 9th bit is added as the parity bit.

Error Detection and Correction

- Errors may occur during communicating data.
- Sending and receiving data may not be the same.
- Different techniques to handle the situation.
- Encoding schemes for error detection and correction.

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Error Detection

- An error detection code can be used to detect errors during transmission.
- The detected error cannot be corrected, but its presence may be indicated.
- Two methods of error detection codes are:
  1. Parity bit coding technique
  2. Check sums technique

Parity

- Error detection scheme.
- Even parity and odd parity.

Even parity

- Each pattern designed to contain even number of 1s.
- If an odd number of 1 encountered it shows an error.

Odd Parity

- Each pattern designed to contain odd number of 1s.
- If an even number of 1s encountered it shows an error.

Parity Checking Examples

Even Parity

Character	Sender	Parity	Receiver	Parity	
E	1000101	**1**	10001011	Even	
A	1000001	**0**	10000010	Even	
q	1110001	**0**	11100000	Odd	Error !!

Odd Parity

Character	Sender	Parity	Receiver	Parity	
E	1000101	**0**	10001010	Odd	
A	1000001	**1**	10000011	Odd	
q	1110001	**1**	11100001	Even	Error !!

d the

Check Sums Technique

- Initially a word "A" is transmitted.
- Another word "B" is also transmitted.
- The binary digits of these two words are added and the sum obtained is retained in the transmitter.
- Again, another word "C" is transmitted and its binary is added to the previous sum retained in the transmitter.
- New sum is retained in the transmitter.
- After transmitting all the words, the final sum called "check sum" is also transmitted.
- The same operation is done at the receiving end and the final sum obtained is being checked against the transmitted check sum.
- There is no error, if the two sums are equal.

Error Correcting Code

- Hamming Code
  - In this code, one or more parity bits are added to a data character in order to detect and correct errors.
- The number of bits changed from one code word to another is known as Hamming Distance.

An error-correcting code

Symbol	Code
A	000000
B	001111
C	010011
D	011100
E	100110
F	101001
G	110101
H	111010